

Patient Identifier:	TEST-VIVA-001	Specimen Type:	Peripheral Blood Smear (WBC)
Staining Protocol:	Wright-Giemsa Cytochemical Stain	Magnification:	100x Oil Immersion Optical Microscopy
Algorithm Engine:	Deep CNN + Grad-CAM Saliency	Clinical Validation:	Automated Pre-Screening Flag

CRITICAL FINDING: ACUTE LYMPHOBLASTIC LEUKEMIA (ALL) DETECTED
Model Classification Confidence: **53.8%** | Morphology Assessment: **Normal N:C Ratio (Mature Lymphocyte)**

1. Multi-Modal Microscopic Cytometry Analysis



(A) Original Smear
Wright-Giemsa 100x View

(B) Grad-CAM Saliency
Nuclear Chromatin Heatmap

(C) Cytoplasm/Nucleus Mask
Morphological Segmentation

2. Computational Morphometric Biomarkers

Cytological Parameter	Measured Value	Reference Normal Range	Clinical Correlation
Nucleus-to-Cytoplasmic (N:C) Ratio	0.325	0.40 - 0.60	NORMAL PHYSIOLOGICAL RANGE
Nuclear Circularity / Regularity	0.847	0.85 - 1.00	Round/Uniform vs. Cleaved/Pleomorphic
Nucleus Pixel Footprint	5,249 px	Standard 45-60 μm^2	Hyperchromatic Blast Nucleus
Cellular Risk Stratification	HIGH (Category 3)	Category 1 (Negative)	Urgent Bone Marrow Biopsy Suggested

3. Interpretive Clinical Observations & Next Steps

Recommendations: High suspicion of Acute Lymphoblastic Leukemia lymphoblasts based on deep neural network activation and elevated N:C ratio. Recommend immediate confirmatory Flow Cytometry (CD10, CD19, TdT immunophenotyping) and Hematopathologist bone marrow aspiration.

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